MILITARY SPECIFICATION SHEET

CHAFF PACKAGE (TYPE RR-173/AL)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the chaff package described herein shall consist of this document and the latest issue of specification MIL-C-85002.

REQUIREMENTS:

Item description: Aluminum foil dipoles.

Detail requirements:

Components: Ten spiral wound dipole assemblies; five Group A and five Group B dipoles (see dimensions herein). Alternate packaging: 10 assemblies containing both group A and B dipoles.

Spiral wound dipole assemblies: Inner case containing dipoles wound spirally on a core in accordance with DL 1050AS113.

Inner case: Constructed of 10-11 point kraft or tagboard and shall conform to DL 1050AS113.

Dipoles:

Material: Plain finish foil, full hard aluminum of 3005 or 5052 alloy.

Thickness: .0018 ± .00018 inch.

Dimensions: Two lengths of dipoles used in chaff packages.

Group A: 27 ± 1.35 inches long and .125 ± .0125 inch wide.

Group B: 13 ± .65 inches long and .125 ± .0125 inch wide.

Coating: None.

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Multiwebb cutting of dipoles: If used, shall not result in metal smear, end welding, or any dipole adhesion.

Core: Plastic and conform to DL 1050AS113.

Winding: 48 +5, -3 Group A or 98 +10, -5 Group B dipoles shall be wound consecutively on core to form a roll 2.3125 ± 0.1156 inches in diameter. An alternate winding shall consist of 24 +3, -2 group A dipoles and 49 +5, -3 group 8 dipoles wound one for two alternately on the same core in the same manner. Dipoles shall be wound to fit inner case and shall not unwind or tangle during storage, but will readily deploy upon dispensing. The free end of the outer dipole shall be taped to the inner case in accordance with DL 1050AS113.

Chaff package: Consists of an outer sleeve containing 10 spiral wound dipole assemblies and vibration protectors in accordance with DL 1050AS113. Of the 10 spiral wound dipole assemblies, 5 shall contain Group A dipoles and 5 shall contain Group B dipoles or ten assemblies that contain both (see dimensions).

Tape marking: Red ink in accordance with DL 1050AS113.

Assembly:

Chaff package: Ten spiral wound dipole assemblies shall be assembled in the chaff package (outer sleeve) with non-chaff components as specified in DL 1050AS113. The package shall separate the individual spirals and prevent unwinding prior to dispensing, but shall allow free and total dispersion upon deployment.

Vibration protectors: Conform to DL 1050AS113.

Dimensions: Are permitted to vary to provide a tight assembled package.

Inner sleeves: Conform to DL 1050AS113.

Quality assurance provisions:

Radar requirements: UHF radar with 1.5 -2 degrees beamwidth horizontally polarized antenna.

Frequency: 435 MHz ± 21.75 MHz

PRF: 320 ± 32 pps

Pulsewidth: 1.0 usec

Radar cross-section test procedures:

Aircraft operation: In accordance with figure 3 of MIL-C-85002 and the flight profile data of table I.

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TABLE I. Flight profile test data.

Flight requirement	Value
Aspect Angle	5° - 20°
Aircraft	180 kts. TAS
Altitude	10,000 feet
Dispense Command	35,000 yards
Standby Command	36,000 yards

Data gathering: For acceptable data in gates 12 through 15, range in yards divided by sine of aspect angle shall be equal to or greater than 211,053.

Standby command: See table I.

Dispense command: See table I.

Data from gates 1 through 15: Spaced 1,000 nanoseconds in time.

Full scale value: $20m^2$

Data processing:

Radar resolution cells: 1000 to 6500 feet behind the aircraft.

Data selection:

Running average: 400 consecutive pulse-to-pulse RCS values.

Correction factors:

Aspect angle: Use ℓ/D ratio of 10 and dipole density of 20 dipoles per meter squared.

Antenna gain: Antenna pattern as specified in MIL-STD-2071 or fourth order polynominal.

Fourth order polynominal for UHF radar:
$0.0249023X^4 + 0.168213X^3 - 0.666504X^2 + 0.0185547X + 0.999268$
(where X = angle off boresite in degrees).

Radar cross-section (RCS): Measured 1000 through 6500 feet behind aircraft shall meet or exceed 5 square meters per gate.

Average of 10 or more units: Equal or exceed 8 square meters per gate.

Preparing Activity
Navy-AS
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